

PRESCRIBED PLURISUBHARMONIC GERMS: HOLOMORPHIC TOWERS AND MASS CONTROL

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ABSTRACT. We study local realization of plurisubharmonic singularities by non-autonomous holomorphic towers. On fixed $\mathbf{C}^n$, $n \geq 2$, an explicit recursion of finite polynomial self-maps realizes every pair $\nu(u, 0) = a$, $(dd^c u)^n = M\delta_0$ with $a \geq 0$, $M > 0$ and $M \geq a^n$. Arbitrary prescribed germs admit exact polynomial towers on varying models, with mixed-mass convergence for asymptotically analytic weights and quantitative estimates for tame weights. For a tame maximal background ϕ , the residual-mass spectrum of its divisorial class is $[\tau(\phi), \infty)$. Exact realization of a prescribed representative on a fixed smooth space fails in general: an explicit tame maximal weight with analytic singularities on $\mathbf{C}^2$ forces all orbit degrees to be at most 16, precluding simultaneous function and mass convergence.

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1. INTRODUCTION AND MAIN RESULTS

The numerical prescription problem has a fixed-space answer: Theorem 7.1 gives one recursive family of finite polynomial self-maps for all $a \geq 0$, $M > 0$, $M \geq a^n$. The data

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determine the ratio of the lower order to the total degree, while a scalar normalization sets the mass. The proof includes $\alpha = 0$ and $\alpha > 0$ in the same construction.

The function prescription problem is to recover a prescribed psh germ ϕ as

$$\phi(z) = \lim_{k \rightarrow \infty} \frac{1}{m_k} \log \|F_{k-1} \circ \cdots \circ F_0(H_0(z))\|.$$

We distinguish three requirements: approximation of a chosen representative, exact holomorphic factorization of the successive approximants, and control of their singularity invariants. Independent Bergman approximants give the first requirement, but do not supply the second. We use the subadditivity of multiplier ideals and the finite algebras $\mathbf{C} + I$ to obtain exact polynomial transitions. Additional singularity estimates then give the third requirement.

The function-prescription construction uses varying embedded models. This flexibility is essential for simultaneous recovery of an arbitrary chosen representative and its residual mass. Theorem 9.1 gives an explicit tame maximal weight on $\mathbf{C}^2$, with analytic singularities and residual mass one, for which every convergent exact finite-map tower on the fixed smooth space has local degrees bounded by 16. Its normalized atomic masses therefore tend to zero when the normalization tends to infinity. The obstruction concerns the chosen representative; realization of a singularity class modulo $O(1)$ is a separate question. The fixed-space prescription of numerical pairs in Theorem 7.1, and the zero-Lelong positive-mass construction of Li–Xia [7], are compatible with this distinction.

Throughout, $\rho \geq 2$ is an integer. We use $d^c = (\partial - \bar{\partial})/(2\pi i)$, so that $dd^c \log |z| = \delta_0$ in one complex dimension. A weight means an isolated psh singularity at 0, with e^ϕ continuous; a maximal weight is maximal off 0. The main approximation theorem also allows non-isolated singularities and does not assume continuity.

Theorem 1.1 (Prescribed scales and exact realization). *Let $\phi \not\equiv -\infty$ be a psh germ on $(\mathbf{C}^n, 0)$ with $\phi(0) = -\infty$. Choose the normalization in either of the following ways:*

(a) m_k is any sequence of positive integers with

$$m_{k+1} \geq \rho m_k, \quad \frac{m_k}{\rho^k} \rightarrow \infty. \quad (1.1)$$

(b) If $\nu(\phi, 0) > 0$, choose an integer m_0 with $\mathcal{J}(m_0\phi)_0$ proper, and put $m_k = m_0\rho^k$. Such m_0 exists.

There are a fixed ball B' centered at 0 and polynomial maps

$$H_k : \mathbf{C}^n \rightarrow \mathbf{C}^{N_k}, \quad F_k : \mathbf{C}^{N_k} \rightarrow \mathbf{C}^{N_{k+1}}$$

such that

- (i) $H_k^{-1}(0) = \{0\}$ and $H_{k+1} = F_k \circ H_k$.
- (ii) Every coordinate of F_k has vanishing order at least ρ .
- (iii) Writing $\mathcal{F}^{[k]} = F_{k-1} \circ \cdots \circ F_0$ and $\mathcal{F}^{[0]} = \text{id}$, one has

$$\phi = \lim_{k \rightarrow \infty} \frac{1}{m_k} \log \|H_k\| = \lim_{k \rightarrow \infty} \frac{1}{m_k} \log \|\mathcal{F}^{[k]} \circ H_0\|. \quad (1.2)$$

Convergence is pointwise, including poles, and in $L^1_{\text{loc}}(B')$. It is locally uniform on every open subset of $B' \setminus \{0\}$ where ϕ is finite and continuous.

- (iv) The images $X_k = H_k(\mathbf{C}^n)$ are closed affine varieties of dimension n . The maps $H_k : \mathbf{C}^n \rightarrow X_k$ are finite normalizations and are isomorphisms off 0. The restrictions $F_k : X_k \rightarrow X_{k+1}$ are finite and birational; their lifts to the common normalization are the identity.
- (v) For every centered divisorial valuation w , with its usual extension to psh germs by generic Lelong numbers,

$$\frac{w((H_k))}{m_k} \rightarrow w(\phi). \quad (1.3)$$

In particular $\nu(m_k^{-1} \log \|H_k\|, 0) \rightarrow \nu(\phi, 0)$.

For instance, $m_k = \rho^k(k+1)$ is available for every target, as are integer sequences $\rho^k \ell_k$ for any nondecreasing $\ell_k \rightarrow \infty$, however slowly. The excess over ρ^k in the zero-Lelong-number case is unavoidable for the required vanishing order.

Proposition 1.2 (Necessary normalization). *Suppose $H_0(0) = 0$, $H_0 \neq 0$, $H_{k+1} = F_k \circ H_k$, and every coordinate of F_k vanishes to order at least ρ . If $m_k^{-1} \log \|H_k\| \rightarrow \phi \neq -\infty$ in L^1_{loc} , and r_0 is the least vanishing order of a nonzero coordinate of H_0 , then*

$$\nu(\phi, 0) \geq r_0 \limsup_{k \rightarrow \infty} \frac{\rho^k}{m_k}.$$

Consequently $\nu(\phi, 0) = 0$ forces $m_k/\rho^k \rightarrow \infty$.

For an isolated singularity ψ , define its mixed residual numbers

$$E_r(\psi) = ((dd^c \psi)^r \wedge (dd^c \log \|z\|)^{n-r})(\{0\}), \quad 1 \leq r \leq n. \quad (1.4)$$

Thus $E_1(\psi) = \nu(\psi, 0)$ and $E_n(\psi) = \tau(\psi)$ is the residual Monge–Ampère mass. Recall that ϕ is *asymptotically analytic* if, for every $0 < \epsilon < 1$, there is an analytic weight θ_ϵ with

$$(1 + \epsilon)\theta_\epsilon + O(1) \leq \phi \leq (1 - \epsilon)\theta_\epsilon + O(1). \quad (1.5)$$

This class includes tame weights [1, 8].

Theorem 1.3 (An exponential tower with mass control). *Let ϕ be a weight with $\nu(\phi, 0) > 0$. For sufficiently large m_0 , put $m_k = m_0 \rho^k$. The tower can be chosen to satisfy*

$$(H_k)\mathcal{O}_{\mathbf{C}^n, 0} = \mathcal{J}(m_k \phi)_0, \quad \mathbf{C}[H_k] = \mathbf{C} + (\mathcal{J}(m_k \phi)_0 \cap \mathbf{C}[z]). \quad (1.6)$$

It has all the conclusions of Theorem 1.1. If ϕ is asymptotically analytic, then, for every $\eta > 0$,

$$\phi + O_k(1) \leq \psi_k \leq (1 - \eta)\phi + O_k(1), \quad \psi_k = m_k^{-1} \log \|H_k\|, \quad (1.7)$$

for all sufficiently large k (it suffices to take $0 < \eta < 1$). Consequently

$$E_r(\psi_k) \rightarrow E_r(\phi), \quad 1 \leq r \leq n.$$

If ϕ is tame with constant C_ϕ , the more precise bounds are

$$\left(1 - \frac{C_\phi}{m_k}\right)^r E_r(\phi) \leq E_r(\psi_k) \leq E_r(\phi), \quad m_k > C_\phi. \quad (1.8)$$

Tameness means that, for some $C_\phi > 0$ and every $t > 0$,

$$f \in \mathcal{J}(t\phi) \implies \log |f| \leq (t - C_\phi)\phi + O(1).$$

This is the constant used in (1.8).

The constants and neighborhoods implicit in $O_k(1)$ may depend on k . The recovery of the chosen representative is proved separately by normalizing the generators on an exhaustion; it is not inferred from these germ inequalities. These constructions do not assume finite generation of a Rees algebra or of a valuation’s associated graded algebra.

The analytic estimates behind mass convergence are classical [1, 8]. The contribution claimed here is their compatibility with the exact factorization and the prescribed normalization. We do not claim new standalone approximation or mixed-multiplicity theorems, nor an independently established literature novelty.

Section 6 gives a further realization by decreasing maximal Green potentials associated with these same orbits. In the tame case their relative error is bounded on the entire punctured domain by an explicit multiple of ρ^{-k} . An ordinary truncation then gives decreasing bounded potentials whose Monge–Ampère measures are supported on shrinking cores. Greenification is an additional envelope operation, and is distinguished from the direct composition limit in (1.2).

Section 8 supplies a complementary construction outside the asymptotically analytic target class. Given an asymptotically analytic background ϕ and a prescribed $M \geq \tau(\phi)$,

the target $\phi + (M - \tau(\phi))^{1/n}h$, with $h = U_0$ from Theorem 7.1, has the same divisorial transform as ϕ and residual mass exactly M . Theorem 8.3 realizes this target by exact polynomial orbits with convergent atomic masses. It combines powered multiplier ideals with powered ideals of the zero-Lelong orbit constructed here; the additional dynamical ideals retain mass that multiplier ideals alone cannot see.

2. PRIMARY IDEALS AND POLYNOMIAL FACTORIZATION

Write $R = \mathcal{O}_{\mathbf{C}^n,0}$, $P = \mathbf{C}[z_1, \dots, z_n]$, $\mathfrak{m} = (z_1, \dots, z_n)R$, and $\mathfrak{n} = (z_1, \dots, z_n)P$.

Lemma 2.1 (Polynomial representatives). *If $I \subset R$ is $\mathfrak{m}$ -primary and $J = I \cap P$, then J is $\mathfrak{n}$ -primary, $JR = I$, and*

$$P/J \simeq R/I. \quad \text{Moreover} \quad J^r = (I^r) \cap P \quad (r \geq 1). \quad (2.1)$$

Proof. Choose q with $\mathfrak{m}^q \subset I$. Taylor truncation of order $q - 1$ shows that every element of R/I is represented by a polynomial. The kernel of this representation is J , and $\mathfrak{n}^q \subset J$. The same Taylor argument writes every element of I as a polynomial in J plus an element of $\mathfrak{m}^q$, proving $JR = I$.

The ideal J^r contains $\mathfrak{n}^{qr}$, so the natural map $P/J^r \rightarrow R/J^r R$ is an isomorphism, again by finite Taylor truncation. Since $J^r R = I^r$, its kernel proves the last assertion. $\square$

Lemma 2.2 (The algebra of a primary ideal). *Let $J \subset P$ be $\mathfrak{n}$ -primary, and put $A = \mathbf{C} + J$. Then A is a finitely generated domain, and J is a maximal ideal of A . For every $r \geq 1$, its power as an ideal of A is, as a subset of P , the polynomial ideal J^r .*

If $\mathfrak{n}^q \subset J$, generators for A can be chosen in J as follows: take polynomial representatives of a vector-space basis of $J/\mathfrak{n}^q$, and all monomials of total degree between q and $2q - 1$.

Proof. The map $c + f \mapsto c$ identifies A/J with $\mathbf{C}$. Any monomial of degree at least $2q$ can have a monomial factor of degree q removed, leaving degree at least q . Induction proves that all monomials of degree at least q lie in the algebra generated by those of degrees $q, \dots, 2q - 1$. Reduction modulo $\mathfrak{n}^q$ then proves finite generation.

One inclusion between the two meanings of J^r is immediate. For the other, absorb a coefficient $p \in P$ in a product $pf_1 \cdots f_r$, $f_i \in J$, into its first factor: $pf_1 \in J \subset A$. Thus every polynomial-ideal product is also an A -ideal product. $\square$

Lemma 2.3 (Finite models). *With $A = \mathbf{C} + J$ as above, P is the finite normalization of A . The induced map $\nu : \mathbf{C}^n \rightarrow \text{Spec } A$ has exactly one point over the point defined by J , and is an isomorphism elsewhere.*

Proof. Each z_i is integral over A , since $z_i^q \in A$. As $P = A[z_1, \dots, z_n]$, this gives finiteness. Furthermore $z_i = z_i^{q+1}/z_i^q$ belongs to $\text{Frac } A$. Thus P and A have the same fraction field, and the normality of P proves the normalization assertion. The vanishing of J in $\mathbf{C}^n$ forces $z_i^q = 0$ for every i , so the fiber over the distinguished point consists only of 0.

For any $h \in J$, the localization A_h equals P_h : indeed, $hp \in J \subset A$ for every $p \in P$. The open sets where such h are nonzero cover the complement of the distinguished point, proving the last assertion. $\square$

Proposition 2.4 (Exact polynomial factorization). *Let $(I_k)_{k \geq 0}$ be $\mathfrak{m}$ -primary ideals of R satisfying*

$$I_{k+1} \subset I_k^\rho \quad (2.2)$$

for an integer $\rho \geq 2$. Put $J_k = I_k \cap P$ and $A_k = \mathbf{C} + J_k$. For each k , choose any finite polynomial tuple H_k with entries in J_k and $\mathbf{C}[H_k] = A_k$. Then $(H_k)R = I_k$, and there are polynomial maps F_k whose coordinates have no terms of degree less than ρ , such that $H_{k+1} = F_k \circ H_k$.

The maps restrict to finite birational morphisms $\text{Spec } A_k \rightarrow \text{Spec } A_{k+1}$, and lift to the identity on $\mathbf{C}^n$.

Proof. Because the entries of H_k vanish at 0 and generate A_k as an algebra, they generate its maximal ideal J_k as an A_k -ideal. Extension to R gives $(H_k)R = I_k$. Lemma 2.1 gives $J_{k+1} \subset J_k^\rho$. An entry h of H_{k+1} is therefore a finite sum of products of ρ elements of J_k . By Lemma 2.2 and the generation assumption, each factor is a polynomial in the entries of H_k . This polynomial can be taken to have zero constant term, since its value at $H_k(0) = 0$ equals the value of the factor at 0. The resulting expression for h is a polynomial in the ambient variables of degree at least ρ .

The inclusion $A_{k+1} \subset A_k \subset P$ realizes the restricted map. The ring P is finite over the Noetherian ring A_{k+1} . Its submodule A_k is consequently finite over A_{k+1} . Both rings have fraction field $\mathbf{C}(z_1, \dots, z_n)$. Finally, all inclusions into P are the literal inclusions, so the induced map on the common normalization is the identity. $\square$

3. ANALYTIC APPROXIMATION WITH ALGEBRA GENERATORS

Shrink the representative of ϕ so that it is defined on a neighborhood of the closure of a ball $B = B_R$, where $R < 1$, and $\phi \leq 0$ on B . This is possible because $\phi(0) = -\infty$. Fix a smaller concentric ball $B' \Subset B$.

For a positive integer m , let $\mathcal{H}_m$ be the Hilbert space of holomorphic functions on B with norm

$$\|f\|_m^2 = \int_B |f|^2 e^{-2m\phi} dV. \quad K_m(z) = \sum_{\ell \geq 1} |f_{m,\ell}(z)|^2, \quad u_m = \frac{1}{2m} \log K_m, \quad (3.1)$$

where $(f_{m,\ell})$ is an orthonormal basis. The kernel K_m is finite and continuous on B ; it is allowed to vanish on the base locus of the Hilbert space. Since $\phi \leq 0$, the weighted unit ball is contained in the ordinary Bergman unit ball, which in particular supplies locally uniform upper bounds.

We use the standard Bergman approximation estimates

$$\phi(z) - \frac{C}{m} \leq u_m(z) \leq \sup_{B(z,r)} \phi + \frac{C - n \log r}{m}, \quad B(z,r) \Subset B. \quad (3.2)$$

They imply pointwise convergence $u_m \rightarrow \phi$, including at poles, and locally uniform convergence on open sets where ϕ is finite and continuous; see [1, Section 5.1, equation (5.1)]. The germs of the basis functions belong to $\mathcal{J}(m\phi)$. The approximation input here is a standard result, rather than a consequence of the tower construction.

For an integer $r \geq 0$, define the normalized homogeneous tuple

$$V_r(z) = \left(\sqrt{\frac{r!}{\beta_1! \cdots \beta_n!}} z^\beta \right)_{|\beta|=r}. \quad \text{Then} \quad \|V_r(z)\|^2 = \|z\|^{2r}, \quad V_0 = (1). \quad (3.3)$$

Lemma 3.1 (A primary polynomial approximation). *Fix integers $m, q \geq 1$ and $s \geq 0$, and set*

$$I = \mathfrak{m}^s(\mathcal{J}(m\phi)_0 + \mathfrak{m}^q), \quad J = I \cap P, \quad A = \mathbf{C} + J.$$

Assume I is proper. For every compact $K \subset B' \setminus \{0\}$ and every $\epsilon > 0$, there is a polynomial tuple H such that

$$\mathbf{C}[H] = A, \quad H^{-1}(0) = \{0\}, \quad (3.4)$$

and

$$\sup_K \left| \frac{1}{m} \log \|H\| - \frac{s}{m} \log \|z\| - \frac{1}{2m} \log (K_m(z) + \|z\|^{2q}) \right| < \epsilon. \quad (3.5)$$

Proof. The expression $T(z) = K_m(z) + \|z\|^{2q}$ is continuous and strictly positive on K . The partial sums of K_m increase uniformly to K_m on K by Dini's theorem. Choose a finite partial sum. Each of its holomorphic basis functions can be uniformly approximated on a

concentric closed ball containing K by a Taylor polynomial p_ℓ . Take the truncation degree at least $q - 1$. The difference of germs belongs to $\mathfrak{m}^q$, so

$$p_\ell \in (\mathcal{J}(m\phi)_0 + \mathfrak{m}^q) \cap P.$$

Consequently the entries of

$$E(z) = V_s(z) \otimes (p_1(z), \dots, p_M(z), V_q(z))$$

belong to J . By taking the partial sum and Taylor degrees large enough, $m^{-1} \log \|E\|$ approximates the target in (3.5) arbitrarily well on K . Strict positivity of T on K justifies passage from an absolute approximation of squared norms to a logarithmic one.

Choose algebra generators $a_1, \dots, a_r$ of A in J as in Lemma 2.2. For sufficiently small $\delta > 0$, appending $(\delta a_1, \dots, \delta a_r)$ to E changes its normalized logarithmic norm on K by less than the remaining error. The resulting tuple H generates A . Its common zero is only 0, either from these generators or directly from $V_s \otimes V_q$. $\square$

Lemma 3.2 (Approximation inside a primary ideal). *Suppose ϕ has an isolated singularity, t, m are positive integers, and I is a proper primary ideal containing $\mathcal{J}(t\phi)_0$. For a compact $K \subset B' \setminus \{0\}$ and $\epsilon > 0$, there is a polynomial tuple H with*

$$\mathbf{C}[H] = \mathbf{C} + (I \cap P), \quad (H)R = I, \quad H^{-1}(0) = \{0\},$$

and

$$\sup_K \left| \frac{1}{m} \log \|H\| - \frac{t}{m} u_t \right| < \epsilon.$$

Proof. The ideal $\mathcal{J}(t\phi)_0$ contains a power of $\mathfrak{m}$: it is nonzero and is the unit ideal off 0, and the assertion is also true if it is the unit ideal at 0. In the present hypotheses that last possibility is excluded by its inclusion in I . High-order monomials in the ideal are integrable against $e^{-2t\phi}$ on B . Local integrability at 0 is ideal membership; away from 0 it follows from boundedness on compact subsets of a slightly larger punctured ball. Thus K_t is positive off 0.

Truncate its basis expansion uniformly on K , then approximate the chosen functions by Taylor polynomials. Taking the degree at least $Q - 1$, where $\mathfrak{m}^Q \subset I$, keeps each Taylor polynomial in I . Append small nonzero multiples of algebra generators of $\mathbf{C} + (I \cap P)$. Positivity of K_t on K gives the required logarithmic estimate, exactly as in Lemma 3.1. The algebra-generation assertion gives $(H)R = I$ and $H^{-1}(0) = \{0\}$. $\square$

We will also use the standard divisorial estimates

$$mw(\phi) - A(w) \leq w(\mathcal{J}(m\phi)_0) \leq mw(\phi). \quad (3.6)$$

Here $w = c \operatorname{ord}_E$ is centered at 0, $w(\phi)$ is the correspondingly scaled generic Lelong number, and $A(w)$ is its log discrepancy. The right inequality follows from the Bergman lower estimate; the left follows by pulling local integrability to a model on which E appears and including the Jacobian factor. This is [1, Section 5.2, equation (5.3)], in nonnegative valuation notation. These values are finite for every fixed centered divisorial valuation and nontrivial psh germ.

4. CONSTRUCTION AT PRESCRIBED SCALES

In case (a) of Theorem 1.1, put $s_k = \rho^k$; in case (b), put $s_k = 0$. In both cases define

$$q_k = m_k^2, \quad I_k = \mathfrak{m}^{s_k} (\mathcal{J}(m_k\phi)_0 + \mathfrak{m}^{q_k}). \quad (4.1)$$

Then

$$s_{k+1} = \rho s_k, \quad q_{k+1} \geq \rho q_k, \quad \frac{s_k}{m_k} \longrightarrow 0, \quad \frac{q_k}{m_k} \longrightarrow \infty.$$

The ideals I_k are primary since they contain $\mathfrak{m}^{s_k + q_k}$. They are proper in case (a) because $s_k \geq 1$, and in case (b) by $\mathcal{J}(m_k\phi)_0 \subset \mathcal{J}(m_0\phi)_0 \subset \mathfrak{m}$. For positive Lelong number, choosing

$0 < c < \nu(\phi, 0)$ gives $\phi \leq c \log \|z\| + O(1)$. An integer m_0 with $m_0 c \geq n$ makes $e^{-2m_0\phi}$ nonintegrable, so $\mathcal{J}(m_0\phi)_0$ is proper.

Lemma 4.1. *The ideals in (4.1) satisfy $I_{k+1} \subset I_k^\rho$.*

Proof. The analytic subadditivity theorem [4] and $m_{k+1} \geq \rho m_k$ give

$$\mathcal{J}(m_{k+1}\phi)_0 \subset \mathcal{J}(\rho m_k\phi)_0 \subset \mathcal{J}(m_k\phi)_0^\rho.$$

Together with the relations for s_k, q_k , this implies

$$I_{k+1} \subset \mathfrak{m}^{\rho s_k} (\mathcal{J}(m_k\phi)_0^\rho + \mathfrak{m}^{\rho q_k}) \subset \mathfrak{m}^{\rho s_k} (\mathcal{J}(m_k\phi)_0 + \mathfrak{m}^{q_k})^\rho = I_k^\rho.$$

□

Proof of Theorem 1.1. Choose increasing compact annuli K_k exhausting $B' \setminus \{0\}$ and $\epsilon_k \downarrow 0$. Lemma 3.1 gives $\mathbf{C}[H_k] = \mathbf{C} + (I_k \cap P)$ and, with $\psi_k = m_k^{-1} \log \|H_k\|$,

$$\sup_{K_k} |\psi_k - W_k| < \epsilon_k, \quad W_k = \frac{s_k}{m_k} \log \|z\| + \frac{1}{2m_k} \log(K_{m_k} + \|z\|^{2q_k}). \quad (4.2)$$

For $z \neq 0$ the elementary log-sum inequality gives

$$\begin{aligned} 0 &\leq W_k - \frac{s_k}{m_k} \log \|z\| \\ &\quad - \max \left\{ u_{m_k}, \frac{q_k}{m_k} \log \|z\| \right\} \leq \frac{\log 2}{2m_k}. \end{aligned} \quad (4.3)$$

Since $B' \subset B_R$ with $R < 1$, the second entry of the maximum tends to $-\infty$ at every nonzero point. The first tends to ϕ by (3.2). The term with s_k/m_k tends to zero. Hence $W_k \rightarrow \phi$ pointwise off 0, including at other poles, and (4.2) gives the same limit for ψ_k . At 0, both ψ_k and ϕ equal $-\infty$. On compact subsets of a region where ϕ is finite and continuous, the Bergman approximation is uniform, ϕ is bounded below, and all the preceding estimates are uniform.

The sequence ψ_k is locally uniformly bounded above also across 0. Indeed, fix a sphere $S_r \subset B' \setminus \{0\}$. For all large k it is contained in K_k . The ordinary Bergman bound for K_{m_k} and $r < 1$ bound W_k uniformly above on S_r , and (4.2) bounds ψ_k there. The maximum principle extends the bound to the interior. The finitely many omitted terms are bounded above separately. Compactness for psh functions and the already established pointwise convergence to a nontrivial psh function imply L^1_{loc} convergence: every L^1 -convergent subsequence has a further almost-everywhere convergent subsequence with limit ϕ , and the alternative of convergence to $-\infty$ locally uniformly is excluded.

Lemma 4.1 and Proposition 2.4 give the exact maps F_k . Lemma 2.3 gives their geometric properties. In particular $(H_k)R = I_k$.

For the valuative assertion, put $a = w(\mathfrak{m}) > 0$. Then

$$w(I_k) = s_k a + \min\{w(\mathcal{J}(m_k\phi)_0), q_k a\}.$$

By (3.6) and $q_k/m_k \rightarrow \infty$, the minimum is the multiplier-ideal term for all sufficiently large k . Consequently

$$w(\phi) + \frac{s_k a - A(w)}{m_k} \leq \frac{w(I_k)}{m_k} \leq w(\phi) + \frac{s_k a}{m_k}. \quad (4.4)$$

This proves (1.3). Taking $w = \text{ord}_0$ gives the Lelong-number assertion. □

Proof of Proposition 1.2. Every monomial in a coordinate of F_k has degree at least ρ . Substitution and possible cancellations therefore give $\text{ord}_0 H_{k+1} \geq \rho \text{ord}_0 H_k$, so $\text{ord}_0 H_k \geq r_0 \rho^k$. Thus

$$\nu(m_k^{-1} \log \|H_k\|, 0) \geq \frac{r_0 \rho^k}{m_k}.$$

Upper semicontinuity of Lelong numbers under L^1_{loc} convergence proves the result. In particular a zero target Lelong number forces $\rho^k/m_k \rightarrow 0$. □

The necessity statement concerns transitions of order at least ρ . It does not apply to systems with surviving linear terms, such as the Li–Xia sequence discussed in Section 10.

5. ISOLATED SINGULARITIES AND RESIDUAL MASSES

Construction in Theorem 1.3. Take m_0 so that $\mathcal{J}(m_0\phi)_0$ is proper. The isolated-singularity hypothesis makes $I_k = \mathcal{J}(m_k\phi)_0$ primary for $m_k = m_0\rho^k$. Subadditivity gives $I_{k+1} \subset I_k^\rho$. Apply Lemma 3.2 with $t = m = m_k$ on an exhaustion of the punctured ball, then use Proposition 2.4. The normalized tuples approximate u_{m_k} on the exhaustion. The pointwise, locally uniform, and L_{loc}^1 convergence follow exactly as in the preceding proof. Their divisorial values satisfy (3.6) after division by m_k . This proves the algebraic and convergence claims, without yet using asymptotic analyticity. $\square$

For a primary ideal I and any tuple generating it, write $\log |I|$ for its logarithmic norm, understood up to a bounded function. If $I \subset J$ then $\log |I| \leq \log |J| + O(1)$. Since $(H_k)R = \mathcal{J}(m_k\phi)_0$,

$$\psi_k = \frac{1}{m_k} \log |\mathcal{J}(m_k\phi)_0| + O_k(1).$$

The following proof makes the uniform multiplicative accuracy of its singularity explicit.

Lemma 5.1. *If ϕ is an asymptotically analytic weight, then for every $0 < \eta < 1$, all sufficiently large integers m satisfy*

$$\phi + O_m(1) \leq \frac{1}{m} \log |\mathcal{J}(m\phi)_0| \leq (1 - \eta)\phi + O_m(1).$$

Proof. The lower bound is the Bergman estimate. Choose $\epsilon > 0$ so small that $(1 - \epsilon)/(1 + \epsilon) > 1 - \eta$, and choose $\theta = \theta_\epsilon$ in (1.5). Analytic weights are tame [1, Lemma 5.10]; let C_θ be a tame constant. Since $\phi \leq (1 - \epsilon)\theta + O(1)$,

$$\mathcal{J}(m\phi)_0 \subset \mathcal{J}(m(1 - \epsilon)\theta)_0.$$

Tameness, applied to a finite set of generators of the ideal on the right, implies

$$\frac{1}{m} \log |\mathcal{J}(m\phi)_0| \leq \left(1 - \epsilon - \frac{C_\theta}{m}\right) \theta + O_m(1).$$

For large m the coefficient is positive. The other inequality in (1.5) gives $\theta \leq \phi/(1 + \epsilon) + O(1)$, and hence

$$\frac{1}{m} \log |\mathcal{J}(m\phi)_0| \leq \frac{1 - \epsilon - C_\theta/m}{1 + \epsilon} \phi + O_m(1) \leq (1 - \eta)\phi + O_m(1)$$

on a negative representative of ϕ . The constants and the neighborhoods may depend on m, η . $\square$

Mass assertions in Theorem 1.3. For weights, the mixed comparison theorem makes E_r invariant under bounded changes and increasing with singularity. Its homogeneity in the first r factors and Lemma 5.1 give, for all sufficiently large k ,

$$(1 - \eta)^r E_r(\phi) \leq E_r(\psi_k) \leq E_r(\phi).$$

First take $k \rightarrow \infty$, then $\eta \downarrow 0$. For tame ϕ , the same argument uses the sharper classical estimate

$$\phi + O_k(1) \leq \psi_k \leq \left(1 - \frac{C_\phi}{m_k}\right) \phi + O_k(1)$$

from [1, Lemma 5.9]. This proves (1.8). The comparison and approximation framework is also developed in [8, Sections 2–4]. $\square$

These inequalities can be read entirely in terms of ideals. For a primary ideal I ,

$$E_r \left(\frac{1}{m} \log |I| \right) = \frac{e(I^{[r]}, \mathbf{m}^{[n-r]})}{m^r}. \quad (5.1)$$

Here brackets denote repetition in mixed multiplicity, not ideal powers. This is the standard local intersection/mixed Monge–Ampère identity; for $r = n$ it reads $\tau(m^{-1} \log |I|) = e(I)/m^n$. One can compute the right side by generic choices of r elements of I and $n - r$ linear forms. See also the mixed-intersection description in [1, Section 4]. Thus Theorem 1.3 controls the complete list of mixed residual numbers through ideals whose generating tuples factor exactly as a holomorphic tower.

6. DECREASING GREEN REALIZATIONS AND SHRINKING CORES

Fix a sufficiently small ball D on which the weights in question are represented. For an isolated singularity u , write

$$\mathcal{G}_D[u] = (\sup\{v \in \text{PSH}^-(D) : v \leq u + O(1) \text{ near } 0\})^*$$

for its complete Greenification. For a primary ideal I , set $G_I = \mathcal{G}_D[\log |I|]$. Analytic singularities admit maximal representatives, and G_I has the same singularity, is maximal off 0, and has boundary value zero. For a continuous maximal weight ϕ the same properties hold for $G_\phi = \mathcal{G}_D[\phi]$, with $G_\phi = \phi + O(1)$ near 0. We use the standard Greenification properties from [8, Sections 2.3 and 4]:

$$\begin{aligned} u \leq v + O(1) &\implies \mathcal{G}_D[u] \leq \mathcal{G}_D[v], & \mathcal{G}_D[cu] &= c\mathcal{G}_D[u] \quad (c > 0), \\ \tau(\mathcal{G}_D[u]) &= \tau(u) \quad \text{for isolated weights.} \end{aligned} \quad (6.1)$$

For continuous isolated weights the envelopes used below are finite off 0 and have the usual zero boundary values.

Theorem 6.1 (Monotone Green envelopes of the orbits). *For the multiplier-ideal tower of an asymptotically analytic weight in Theorem 1.3, the functions*

$$P_k = \frac{G_{\mathcal{J}(m_k \phi)_0}}{m_k} = \mathcal{G}_D \left[\frac{1}{m_k} \log \|\mathcal{F}^{[k]} \circ H_0\| \right]$$

decrease to G_ϕ . For every $\eta > 0$, eventually

$$G_\phi \leq P_k \leq (1 - \eta)G_\phi \quad (0 < \eta < 1).$$

Thus $P_k/G_\phi \rightarrow 1$ uniformly on $D \setminus \{0\}$. If ϕ is tame, one can use $\eta = C_\phi/m_k$.

P_k is maximal off 0, and $(dd^c P_k)^n = \tau(P_k)\delta_0$ with $\tau(P_k) \rightarrow \tau(\phi)$.

Proof. For the multiplier tower, subadditivity and $m_{k+1} = \rho m_k$ give

$$G_{\mathcal{J}(m_{k+1} \phi)_0} \leq G_{\mathcal{J}(m_k \phi)_0}^\rho = \rho G_{\mathcal{J}(m_k \phi)_0}.$$

Hence $P_{k+1} \leq P_k$. Applying (6.1) to Lemma 5.1 gives $G_\phi \leq P_k \leq (1 - \eta)G_\phi$ for large k . These global inequalities identify the decreasing limit and give the ratio convergence. The tame estimate follows in the same way from the quantitative singularity sandwich.

Maximality and the support of the measures are properties of the Green envelopes. The mass convergence follows from Theorem 1.3 and (6.1). $\square$

Corollary 6.2 (Decreasing bounded truncations). *For the sequence in Theorem 6.1, choose any increasing $T_k \rightarrow \infty$, with $T_k > 0$, and put*

$$V_k = \max\{P_k, -T_k\}, \quad K_k = \{P_k \leq -T_k\}.$$

Then $V_k \downarrow G_\phi$, the sets K_k shrink to 0 in the sense that they eventually lie in every neighborhood of 0, and

$$(dd^c V_k)^n \rightharpoonup \tau(\phi)\delta_0.$$

More precisely, $(dd^c V_k)^n$ is supported on $\{P_k = -T_k\}$ and has total mass $\tau(P_k)$.

Proof. Both entries in the maximum decrease with k , hence V_k decreases, and its limit is G_ϕ by Theorem 6.1. Since $P_k \geq G_\phi$,

$$K_k \subset \{G_\phi \leq -T_k\}.$$

The isolated singularity and zero boundary behavior of G_ϕ make the sets on the right compact in D and force them to shrink to 0. The analytic Green weights P_k are continuous off 0 and tend to $-\infty$ at 0. Demailly's sweeping formula [3, Chapter III, (6.1)–(6.2)], gives the support and total-mass assertions. For any compactly supported continuous χ on D ,

$$\left| \int_D \chi (dd^c V_k)^n - \tau(P_k) \chi(0) \right| \leq \tau(P_k) \sup_{K_k} |\chi - \chi(0)| \longrightarrow 0.$$

Together with $\tau(P_k) \rightarrow \tau(\phi)$ this proves the measure limit. $\square$

The corollary imports the monotone-truncation and shrinking-core mechanism into the prescribed-weight construction. Its inputs are the exact orbit ideals, followed by their Green envelopes. It does not assert that Greenification itself is a finite-dimensional holomorphic map.

7. ONE FINITE-MAP RECURSION FOR EVERY LELONG-MASS PAIR

Here the state space is the fixed smooth space $\mathbf{C}^n$, $n \geq 2$. The prescribed data determine the orders in a single recursive family of polynomial maps. No zero-Lelong positive-mass weight is assumed as an input. The perturbation and truncation mechanism is inspired by Li-Xia [7]; the proof below includes its zero-Lelong endpoint, together with every intermediate Lelong number, in the same algorithm. The map degrees are allowed to grow.

Theorem 7.1 (A uniform recursion on a fixed smooth space). *For every $\alpha \in [0, 1]$, the recursion below constructs finite polynomial self-maps F_j of $\mathbf{C}^n$ and $G_j = F_j \circ \dots \circ F_1$ such that, on the unit polydisc $\mathbb{D}^n$,*

$$q_j = \frac{1}{N_j} \log \|G_j\|_\infty \longrightarrow U_\alpha, \quad N_j = 2^{j(j+1)/2}.$$

The convergence is pointwise, locally uniform off 0, and in L^1_{loc} . The function U_α has an isolated pole, e^{U_α} is continuous, and

$$\nu(U_\alpha, 0) = \alpha, \quad (dd^c U_\alpha)^n = \delta_0. \quad (7.1)$$

At every finite stage,

$$\nu(q_j, 0) = \alpha_j \longrightarrow \alpha, \quad (dd^c q_j)^n = \delta_0. \quad (7.2)$$

Consequently every finite pair $a \geq 0$, $M > 0$ with $M \geq a^n$ is realized by this same recursion, with

$$\lambda = M^{1/n}, \quad \alpha = \frac{a}{\lambda}, \quad u = \lambda U_\alpha = \lim_j \frac{\lambda}{N_j} \log \|G_j\|_\infty, \quad (dd^c u)^n = M \delta_0, \quad \nu(u, 0) = a. \quad (7.3)$$

7.1. The complete parameter recursion. Start with

$$N_0 = P_0 = \alpha_0 = 1, \quad T_0 = \gamma_0 = \kappa_0 = 0, \quad G_0(z) = z.$$

For $j \geq 1$, define, in the following order,

$$D_j = 2^j, \quad p_j = \max \left\{ 1, \left\lceil \frac{D_j \alpha}{\alpha_{j-1}} \right\rceil \right\}, \quad (7.4)$$

$$N_j = D_j N_{j-1}, \quad P_j = p_j P_{j-1}, \quad \alpha_j = P_j / N_j, \quad (7.5)$$

$$\varepsilon_j = \frac{1}{8} \exp(-(D_j - p_j) N_{j-1} T_{j-1}), \quad (7.6)$$

$$\gamma_j = p_j \gamma_{j-1} - \log \varepsilon_j, \quad \kappa_j = \frac{D_j}{p_j} \kappa_{j-1} + \frac{\log 8}{N_j}, \quad (7.7)$$

$$L_j = j \left(1 + \frac{\gamma_j}{N_j} \right), \quad (7.8)$$

$$T_j = \max \left\{ T_{j-1} + 4, 1 + \alpha_j L_j + \frac{\gamma_j}{N_j}, 1 + \frac{j}{\alpha_j} + \kappa_j \right\}. \quad (7.9)$$

All these numbers are finite and explicit; no equation involving an unknown psh limit has to be solved. Define

$$F_j(z_1, z_2, z_3, \dots, z_n) = \left(\frac{1}{4} z_2^{D_j}, \frac{1}{4} z_1^{D_j} + \varepsilon_j z_2^{p_j}, \frac{1}{4} z_3^{D_j}, \dots, \frac{1}{4} z_n^{D_j} \right). \quad (7.10)$$

The extra coordinates are omitted when $n = 2$. The choice (7.4) gives $1 \leq p_j \leq D_j$, $0 < \alpha_j \leq \alpha_{j-1}$ and

$$0 \leq \alpha_j - \alpha \leq D_j^{-1} = 2^{-j}. \quad (7.11)$$

For $\alpha = 0$ it gives $p_j = 1$ at every stage; for $\alpha = 1$ it gives $p_j = D_j$. These are endpoints of the same formula.

7.2. Degree, order, and coercivity. Each F_j is proper and finite of degree D_j^n , with $F_j^{-1}(0) = \{0\}$: its first coordinate determines z_2 by a monic equation of degree D_j , its second then determines z_1 , and each remaining coordinate is another degree- D_j equation. Thus $\deg_0 G_j = N_j^n$ and $G_j^{-1}(0) = \{0\}$.

Write a_j, b_j for the orders of the first and second components of G_j . All polynomial coefficients are nonnegative real numbers, so leading terms cannot cancel. Starting with $a_0 = b_0 = 1$, induction gives

$$a_j = D_j P_{j-1}, \quad b_j = P_j.$$

Indeed $a_{j-1} \geq b_{j-1}$, so $\min\{D_j a_{j-1}, p_j b_{j-1}\} = p_j b_{j-1}$. The other components have order N_j . Hence $\text{ord}_0 G_j = P_j$ and $\nu(q_j, 0) = P_j / N_j$.

We will use the elementary estimate, for $0 \leq r = \|z\|_\infty \leq 1$,

$$\frac{1}{8} r^{D_j^2/p_j} \leq \|F_j(z)\|_\infty \leq \frac{3}{8} r^{p_j} \leq \frac{3}{8} r. \quad (7.12)$$

For the lower bound, if a coordinate other than z_1 has modulus r , its pure-power output is $r^{D_j}/4$. Otherwise $|z_1| = r$. If $|z_2|^{p_j} \leq r^{D_j}$, the second output has modulus at least $r^{D_j}/8$; if $|z_2|^{p_j} > r^{D_j}$, the first output exceeds $r^{D_j^2/p_j}/4$. Here $\varepsilon_j \leq 1/8$ and $D_j^2/p_j \geq D_j$ were used. The upper bound is immediate.

Iterating the lower bound gives, by (7.7),

$$q_j(z) \geq \alpha_j^{-1} \log \|z\|_\infty - \kappa_j. \quad (7.13)$$

In particular, the compact cutoff cores

$$K_j = \{z \in \overline{\mathbb{D}^n} : q_j(z) \leq -T_j\} \subset \{\|z\|_\infty < e^{-j}\} \quad (7.14)$$

shrink to the origin. The strict inclusion follows from the last term of (7.9).

For completeness, the standard finite-map mass formula gives

$$(dd^c \log \|G_j\|_\infty)^n = N_j^n \delta_0. \quad (7.15)$$

One way to see the formula is to truncate $\log \|w\|_\infty$ at level $-A$, apply change of variables for the finite map, and let $A \rightarrow \infty$. The max-norm mass is δ_0 , and its pullback has support $G_j^{-1}(0) = \{0\}$ and mass N_j^n . The necessary monotone continuity and the max-norm computation are classical; see [3, Chapter III, Sections 3–4 and 7]. Dividing (7.15) by N_j^n proves the mass identity in (7.2).

7.3. One-step comparison and the limit. If $q_j(z) \geq -T_j$, put $r = \|G_j(z)\|_\infty$. Then $r \geq e^{-N_j T_j}$, and (7.6) implies

$$\varepsilon_{j+1} r^{p_{j+1}} \leq \frac{1}{8} r^{D_{j+1}}.$$

The unperturbed output has norm $r^{D_{j+1}}/4$. The triangle and reverse triangle inequalities therefore give

$$-\frac{\log 8}{N_{j+1}} \leq q_{j+1} - q_j \leq 0 \quad \text{on } \{q_j \geq -T_j\}. \quad (7.16)$$

If instead $q_j(z) < -T_j$, comparison with $r_0 = e^{-N_j T_j}$ gives

$$\|F_{j+1}(G_j(z))\|_\infty \leq \frac{1}{4} r_0^{D_{j+1}} + \varepsilon_{j+1} r_0^{p_{j+1}} \leq \frac{3}{8} r_0^{D_{j+1}}.$$

Thus $q_{j+1} < -T_j$ in that case. Since $T_{j+1} \geq T_j + 4$, the continuous bounded psh functions

$$V_j = \max\{q_j, -T_j\}$$

satisfy $V_{j+1} \leq V_j$ everywhere.

Moreover, $q_j(z) \geq -T_j$ implies $q_{j+1}(z) > -T_{j+1}$ by (7.16); this implication persists at every later index. Summing the one-step bounds gives

$$0 \leq q_j - q_k \leq \log 8 \sum_{l=j+1}^k \frac{1}{N_l} \leq \frac{\log 8}{N_j} \quad (k > j) \quad \text{on } \{q_j \geq -T_j\}. \quad (7.17)$$

By (7.14), every punctured compact set is in these regions for all sufficiently large j . Thus q_j is uniformly Cauchy there, with a finite continuous limit. The same limit equals $U_\alpha = \lim V_j$, so it is negative and psh. At 0, $V_j(0) = -T_j \rightarrow -\infty$. Given any $A > 0$, choose $T_j > A$; continuity of V_j near 0 and $U_\alpha \leq V_j$ show that $U_\alpha(z) \rightarrow -\infty$ as $z \rightarrow 0$. This proves continuity of e^{U_α} and the isolated-pole assertion. The q_j are locally uniformly bounded above by zero; psh compactness and convergence off 0 also give convergence in L_{loc}^1 .

Demailly's sweeping formula [3, Chapter III, (6.1)–(6.2)] applies to q_j , because (7.13) and (7.9) put its cutoff sublevel compactly inside the polydisc. It says that $(dd^c V_j)^n$ is a probability measure supported on $\{q_j = -T_j\} \subset K_j$. Consequently $(dd^c V_j)^n \rightarrow \delta_0$. On the other hand, $V_j \downarrow U_\alpha$ and the limit has only an isolated unbounded point. Monotone continuity [3, Chapter III, Corollary 4.3] identifies this weak limit with $(dd^c U_\alpha)^n$. This proves its asserted mass and maximality off 0.

7.4. The prescribed Lelong number of the limit. It is not enough to take the limit of $\nu(q_j, 0)$. Upper semicontinuity under L_{loc}^1 convergence supplies only $\nu(U_\alpha, 0) \geq \alpha$. For the other direction consider the positive real ray re_2 . The second component satisfies

$$(G_j(re_2))_2 \geq e^{-\gamma_j} r^{P_j}, \quad 0 < r < 1,$$

by induction using the nonnegative coefficients and (7.7). At $r_j = e^{-L_j}$ this gives

$$q_j(r_j e_2) \geq -\alpha_j L_j - \frac{\gamma_j}{N_j} \geq -T_j + 1.$$

All subsequent cutoffs are therefore inactive at this point, and (7.17) yields

$$\frac{U_\alpha(r_j e_2)}{\log r_j} \leq \alpha_j + \frac{\gamma_j}{N_j L_j} + \frac{\log 8}{N_j L_j} \leq \alpha_j + \frac{1}{j} + \frac{\log 8}{N_j L_j}.$$

Since $L_j \geq j$, the points tend to 0 and the right side tends to α . The liminf characterization of the Lelong number proves $\nu(U_\alpha, 0) \leq \alpha$. This completes the proof of Theorem 7.1.

If integer normalization parameters are desired in (7.3), take $m_j = \lceil N_j/\lambda \rceil$. Then $N_j/m_j \rightarrow \lambda$, and $m_j^{-1} \log \|G_j\|_\infty$ has the same limit with convergent Lelong numbers and atomic masses. The degenerate pair $(0, 0)$ is realized by the constant psh function zero; if an actual pole is required, the radial weight $-\sqrt{-\log \|z\|}$ and Theorem 1.1(a) give a zero-mass isolated pole and an exact varying-model tower. This degenerate pole assertion is separate from the positive-mass fixed-space construction above.

The construction uses one parameter recursion for $\alpha = 0$, $0 < \alpha < 1$, and $\alpha = 1$. Its degrees $D_j = 2^j$ are unbounded, and the coefficients can be extremely small. It makes no claim about bounded-degree or autonomous realizations. Its proof includes a counterexample to the zero-mass conjecture without invoking the existence theorem of [7]; the adaptation of their perturbation-and-sweeping strategy remains part of the attribution.

8. PRESCRIBING MASS WITH FIXED DIVISORIAL DATA

Throughout this section $n \geq 2$, $\ell(z) = \log \|z\|$, and all Monge–Ampère products are the classical products for isolated singularities, equivalently their local Cegrell products. In particular they include the mass at the pole. Fix the zero endpoint $h = U_0$ of Theorem 7.1, which satisfies

$$\nu(h, 0) = 0, \quad (dd^c h)^n = \delta_0. \quad (8.1)$$

Its existence and all the required orbit estimates were proved in Section 7. The construction below applies the same zero endpoint to arbitrary prescribed backgrounds.

8.1. A deformation with an exact mass formula. For isolated psh germs put

$$m(u_1, \dots, u_n) = (dd^c u_1 \wedge \dots \wedge dd^c u_n)(\{0\}).$$

This is finite, nonnegative, symmetric and multilinear. We write brackets for repetition. The reversed Alexandrov–Fenchel inequality of Kim–Rashkovskii [5, Theorem 1.4] says that

$$m(u, v, w_3, \dots, w_n)^2 \leq m(u, u, w_3, \dots, w_n)m(v, v, w_3, \dots, w_n). \quad (8.2)$$

The local formulation applies to isolated psh singularities after adding smooth functions, which does not change these point masses.

Lemma 8.1. *The seed satisfies $E_j(h) = 0$ for $1 \leq j < n$. If $v_1, \dots, v_{n-j}$ have isolated singularities and $v_i \geq L_i \ell + O(1)$ for some finite $L_i > 0$, then*

$$m(h^{[j]}, v_1, \dots, v_{n-j}) = 0, \quad 1 \leq j < n.$$

For a centered divisorial valuation w , one also has $w(h) = 0$.

Proof. The sequence $E_j(h)$ is log-convex by (8.2), and $E_1(h) = 0$, $E_n(h) = 1$. The inequality $E_2(h)^2 \leq E_1(h)E_3(h)$ gives $E_2(h) = 0$ when this index occurs; induction gives all the remaining indices below n . Mixed comparison bounds the displayed mixed mass from above by $\prod_i L_i E_j(h) = 0$. Finally Skoda integrability gives $\mathcal{J}(mh)_0 = R$ for every $m > 0$. The divisorial approximation estimate $mw(h) - A(w) \leq w(\mathcal{J}(mh)) = 0$ and $m \rightarrow \infty$ give the last assertion. $\square$

Theorem 8.2 (Mass deformation over an arbitrary background). *Let u be a psh germ with an isolated singularity and put $u_t = u + th$ for $t \geq 0$. Then u_t has the same divisorial transform as u . The function $t \mapsto \tau(u_t)$ is a continuous strictly increasing map from $[0, \infty)$ onto $[\tau(u), \infty)$.*

If in addition $u \geq L\ell + O(1)$ for a finite $L > 0$, then

$$\tau(u_t) = \tau(u) + t^n. \quad (8.3)$$

More generally, for every isolated reference Ψ with $\Psi \geq L_\Psi \ell + O(1)$,

$$m(u_t^{[r]}, \Psi^{[n-r]}) = m(u^{[r]}, \Psi^{[n-r]}), \quad 1 \leq r < n. \quad (8.4)$$

Locally bounded backgrounds, including $u = 0$, are allowed as well.

Proof. Divisorial Lelong numbers are additive, so Lemma 8.1 gives $w(u_t) = w(u)$. Multilinearity gives the finite polynomial

$$\tau(u_t) = \sum_{j=0}^n \binom{n}{j} t^j m(u^{[n-j]}, h^{[j]}). \quad (8.5)$$

All coefficients are nonnegative and the coefficient of t^n is one. Thus the polynomial is continuous and strictly increasing on $[0, \infty)$ and tends to infinity. Under the logarithmic lower bound, every intermediate coefficient vanishes by Lemma 8.1. The same argument, after expanding the r factors, proves (8.4). $\square$

Every isolated asymptotically analytic weight has the logarithmic lower bound in this theorem: take one analytic comparison weight in (1.5) and apply its Łojasiewicz inequality. In particular this covers every tame maximal weight, including those representing valuations.

For such a weight ϕ and any $M \geq \tau(\phi)$ the formula

$$u_M = \phi + (M - \tau(\phi))^{1/n} h \quad (8.6)$$

therefore prescribes the mass exactly, while preserving the divisorial transform and all proper mixed numbers in (8.4). The deformed germ is not asserted to remain maximal or tame. The failure of residual mass to be a valuative invariant already occurs in earlier work of Chi Li [6]. The uniform construction of Section 7 supplies the zero-Lelong summand here.

8.2. Exact towers whose atomic masses also converge. The preceding formula alone would only prescribe a psh function. We next make it compatible with the polynomial factorization of this paper, including convergence of the masses at the origin.

Let $B_k : \mathbf{C}^n \rightarrow \mathbf{C}^n$ be the composition G_k in Theorem 7.1 at $\alpha = 0$, and retain $N_k = 2^{k(k+1)/2}$ and $D_k = 2^k$. Thus

$$\begin{aligned} B_{k+1} &= F_{k+1} \circ B_k, & B_k^{-1}(0) &= \{0\}, \\ q_k &= N_k^{-1} \log \|B_k\| \longrightarrow h, & \nu(q_k, 0) &= N_k^{-1}, \quad \tau(q_k) = 1. \end{aligned} \quad (8.7)$$

Convergence is locally uniform off 0 and in L_{loc}^1 . Changing between Euclidean and maximum norms adds $O(N_k^{-1})$.

Theorem 8.3 (Mass-preserving towers with a zero-Lelong summand). *Let ϕ be an isolated asymptotically analytic weight, $t \geq 0$, and $\rho \geq 2$ an integer. There are polynomial tuples H_k and polynomial transitions F_k , all vanishing at 0, such that*

$$H_{k+1} = F_k \circ H_k, \quad \text{ord}_0 F_k \geq \rho, \quad \chi_k = M_k^{-1} \log \|H_k\| \longrightarrow \phi + th, \quad M_k = m_0 N_k \rho^k. \quad (8.8)$$

One may discard finitely many initial indices. Convergence is pointwise, locally uniform off 0, and in L_{loc}^1 . The orbit ideals satisfy

$$(H_k)R = \mathcal{J}(m_0 N_k \phi)_0^{\rho^k} ((B_k)R)^{\lfloor tm_0 \rho^k \rfloor}. \quad (8.9)$$

All the mixed residual numbers converge:

$$E_r(\chi_k) \longrightarrow E_r(\phi) \quad (1 \leq r < n), \quad \tau(\chi_k) \longrightarrow \tau(\phi) + t^n. \quad (8.10)$$

For $\phi = 0$ and $t > 0$ the same statements hold after replacing the first ideal by R and taking $m_0 = 1$. The models have intrinsic dimension n ; their ambient dimensions may vary as in Theorem 1.1.

Proof. Use the construction in the proof of Theorem 1.3 at levels $a_k = m_0 N_k$. Its proof works verbatim for these levels: subadditivity gives $\mathcal{J}(a_{k+1}\phi) \subset \mathcal{J}(a_k\phi)^{D_{k+1}}$, and Lemma 3.2 normalizes the generators at each level. All analytic and mass estimates depend only on $a_k \rightarrow \infty$. Write the resulting tuple as A_k and put $p_k = a_k^{-1} \log \|A_k\|$. With $P = \mathbf{C}[z]$, put

$$\mathbf{a}_k = (A_k)P, \quad \mathbf{b}_k = (B_k)P, \quad R_k = \rho^k, \quad s_k = \lfloor tm_0 R_k \rfloor, \quad t_k = \frac{s_k}{m_0 R_k}.$$

Both polynomial ideals are primary at 0. Moreover $\mathfrak{a}_{k+1} \subset \mathfrak{a}_k^{D_{k+1}}$, $\mathfrak{b}_{k+1} \subset \mathfrak{b}_k$, and $s_{k+1} \geq \rho s_k$. Therefore

$$\mathfrak{k}_k = \mathfrak{a}_k^{R_k} \mathfrak{b}_k^{s_k} \implies \mathfrak{k}_{k+1} \subset \mathfrak{k}_k^\rho. \quad (8.11)$$

Here $0 \leq t - t_k < (m_0 \rho^k)^{-1}$.

Let P_k be the tensor tuple $A_k^{\otimes R_k} \otimes B_k^{\otimes s_k}$, with the zeroth tensor power interpreted as the scalar 1. Its entries generate $\mathfrak{k}_k$, and its Euclidean norm satisfies

$$\frac{1}{M_k} \log \|P_k\| = p_k + t_k q_k, \quad M_k = a_k R_k. \quad (8.12)$$

Choose a tuple Q_k generating the finite algebra $\mathbf{C} + \mathfrak{k}_k$, with entries in $\mathfrak{k}_k$. Appending $\delta_k Q_k$ for sufficiently small $\delta_k > 0$ gives $H_k = (P_k, \delta_k Q_k)$ with $\mathbf{C}[H_k] = \mathbf{C} + \mathfrak{k}_k$ and $(H_k)P = \mathfrak{k}_k$. Choose δ_k so that the normalized logarithms of H_k and P_k differ by at most $1/k$ on the k th compact annulus of a fixed exhaustion. This is possible because P_k never vanishes away from 0. Also arrange $\|\delta_k Q_k\| \leq 1$ on a fixed closed ball. This gives a uniform local upper bound for the normalized logarithms. Their pointwise convergence off 0, and their common pole at 0, imply the claimed L^1_{loc} convergence by psh compactness. Proposition 2.4, applied to (8.11), gives the exact transitions of order at least ρ . The same finite-algebra argument gives the stated intrinsic dimension and normalization properties.

It remains to check atomic masses; the appended coordinates are a bounded change of singularity at each fixed level. By (8.2) and (8.7),

$$0 \leq E_j(q_k) \leq N_k^{-(n-j)/(n-1)}, \quad 1 \leq j < n. \quad (8.13)$$

Indeed this is the interpolation inequality for the log-convex sequence between its indices 1 and n . Choose L with $\phi \geq L\ell + O(1)$. The lower Bergman estimate gives $p_k \geq \phi + O_k(1)$, so mixed comparison gives

$$m(p_k^{[n-j]}, q_k^{[j]}) \leq L^{n-j} E_j(q_k), \quad 1 \leq j < n.$$

Consequently the actual atomic masses obey the explicit bound

$$0 \leq \tau(\chi_k) - \tau(p_k) - t_k^n \leq \sum_{j=1}^{n-1} \binom{n}{j} t_k^j L^{n-j} N_k^{-(n-j)/(n-1)}. \quad (8.14)$$

Since $\tau(p_k) \rightarrow \tau(\phi)$, this proves the top-degree claim. For $r < n$, expand $E_r(p_k + t_k q_k)$ and bound each term with $j \geq 1$ copies of q_k by $L^{r-j} E_j(q_k)$. These terms vanish by (8.13); the term with $j = 0$ converges by Theorem 1.3. Theorem 8.2 identifies these limits with the invariants of $\phi + th$. For $\phi = 0$, take $A_k = (1)$ and retain the remaining construction. $\square$

The same comparison proof works with ℓ replaced in the $n - r$ reference factors by any fixed isolated weight with a finite logarithmic lower bound. Thus the construction also recovers the corresponding proper mixed numbers of the target. It does not use continuity of atomic mass under L^1 convergence; (8.14) supplies the missing control directly.

8.3. The mass spectrum of a divisorial class.

Corollary 8.4 (The mass spectrum of a tame valutive class). *Fix a tame maximal weight ϕ . Among isolated psh germs u with $w(u) = w(\phi)$ for every centered divisorial valuation w , the set of residual masses is exactly*

$$[\tau(\phi), \infty).$$

Every mass in this interval is realized by (8.6) and by a tower satisfying Theorem 8.3.

Proof. If u has the given transform, [1, Theorem 5.12] implies $u \leq \phi + O(1)$. Comparison then gives $\tau(u) \geq \tau(\phi)$. Theorem 8.2 supplies every value above this lower endpoint, with the required transform. The tower follows from Theorem 8.3. $\square$

9. AN OBSTRUCTION ON A FIXED SMOOTH STATE SPACE

Exact recovery of a prescribed representative and its residual mass by finite towers on one fixed smooth space fails in general. The counterexample already occurs in dimension two and has analytic singularities. Its obstruction is fiber invariance: every identification made at one stage persists in the limiting function.

Put

$$P(z_1, z_2) = |z_1|^4 + |z_2|^4 + |z_1 z_2|^2 + \frac{1}{2} \operatorname{Re}(z_1^2 \bar{z}_2^2), \quad (9.1)$$

$$\phi(z) = \frac{1}{4} \log P(z), \quad \phi(0) = -\infty. \quad (9.2)$$

Here and below the norm of a holomorphic tuple is Euclidean unless otherwise stated.

Theorem 9.1 (A fixed smooth-space obstruction). *The function ϕ is a tame maximal weight with analytic singularities, and*

$$\nu(\phi, 0) = 1, \quad (dd^c \phi)^2 = \delta_0.$$

Let $G_k : (\mathbf{C}^2, 0) \rightarrow (\mathbf{C}^2, 0)$ be finite holomorphic map germs with representatives holomorphic on one fixed neighborhood of 0. Suppose that there are holomorphic map germs F_k with

$$G_{k+1} = F_k \circ G_k$$

as germs, and positive real numbers m_k , such that

$$q_k = \frac{1}{m_k} \log \|G_k\| \longrightarrow \phi$$

pointwise on that neighborhood, or in L_{loc}^1 . Then

$$\deg_0 G_k \leq 16 \quad \text{for every } k. \quad (9.3)$$

Consequently, if $m_k \rightarrow \infty$, then

$$\tau(q_k) = \frac{\deg_0 G_k}{m_k^2} \longrightarrow 0 \neq \tau(\phi).$$

Thus universal simultaneous exact factorization, function recovery, and mass convergence fail even for tame maximal weights with analytic singularities, already in dimension two.

The theorem requires no bound on the degrees or coefficients of the transition maps, no lower bound on their vanishing orders, and no polynomial assumption. For $m_k \rightarrow \infty$ its conclusion is unchanged if any other fixed equivalent norm is used. Section 9.4 also rules out mass convergence with bounded normalizations in the Euclidean-norm formulation.

9.1. The analytic maximal weight. Define the full quadratic tuple

$$H(z) = \left(\sqrt{\frac{5}{8}}(z_1^2 + z_2^2), \sqrt{\frac{3}{8}}(z_1^2 - z_2^2), z_1 z_2 \right).$$

Direct expansion gives

$$P = \|H\|^2, \quad \phi = \frac{1}{2} \log \|H\|. \quad (9.4)$$

The three components of H form a basis of the quadratic homogeneous polynomials in two variables. In particular they generate $\mathfrak{m}^2$, and $H^{-1}(0) = \{0\}$. Moreover, P is positive off zero and homogeneous of bidegree $(2, 2)$. Compactness of the unit sphere therefore gives

$$\phi = \log \|z\| + O(1), \quad \phi(tz) = \log |t| + \phi(z) \quad (t \in \mathbf{C}^*). \quad (9.5)$$

It follows from (9.4) that ϕ is psh and has analytic singularities. The function $e^\phi = P^{1/4}$ is continuous at zero and smooth elsewhere.

The projective map

$$\mathbf{P}^1 \longrightarrow \mathbf{P}^2, \quad [z] \longmapsto [H(z)]$$

is a Veronese embedding followed by an invertible projective linear map. Thus $dd^c\phi$, away from zero, has rank exactly one: its kernel is the complex radial line, and its induced form in the projective direction is strictly positive. In particular $(dd^c\phi)^2 = 0$ off zero. Bounded equivalence in (9.5) and the local comparison theorem give $\nu(\phi, 0) = \tau(\phi) = 1$.

For completeness, tameness follows directly with the BFJ definition [1, Definition 5.8]. If $f \in \mathcal{J}(t\phi)$ is nonzero and $p = \text{ord}_0 f$, bounded equivalence to $\log \|z\|$ and radial integration in $\mathbf{C}^2$ give

$$p > t - 2.$$

Since $\log |f| \leq p \log \|z\| + O(1)$, on a sufficiently small ball this implies

$$\log |f| \leq (t - 2)\phi + O(1).$$

Hence $C_\phi = 2$ is a tame constant.

9.2. Fiber invariance and linearity of local symmetries.

Lemma 9.2. *Under the factorization and convergence assumptions of Theorem 9.1, the limit ϕ is constant on the local fibers of each fixed G_j .*

Proof. Fix j . For every $k \geq j$, the germ G_k factors through G_j . Therefore $G_k(z) = G_k(w)$ on the germ at $(0, 0)$ of

$$Z_j = \{(z, w) : G_j(z) = G_j(w)\}.$$

This germ has a representative with finitely many irreducible components, all passing through $(0, 0)$. Take this representative inside the product of the fixed common domain of the G_k . For each k , the identity theorem on every component extends the germ equality throughout this representative. Thus the neighborhood on which fiber equality holds can be chosen independently of k . This argument is relevant when the transitions are only germs, rather than globally defined polynomials.

On the regular locus of G_j , let σ be a local inverse branch correspondence: it is a biholomorphism between two source open sets with $G_j \circ \sigma = G_j$. Then

$$q_k \circ \sigma = q_k \quad (k \geq j).$$

Pointwise convergence gives $\phi \circ \sigma = \phi$. For L^1_{loc} convergence the same conclusion holds: composition with a fixed local biholomorphism preserves local L^1 convergence on relatively compact subdomains, by the change-of-variables formula. Equality almost everywhere is equality everywhere here because ϕ is continuous off zero. Continuity then also gives the statement on the exceptional fibers. $\square$

Lemma 9.3. *Suppose σ is a biholomorphism between nonempty connected open subsets of $\mathbf{C}^2 \setminus \{0\}$, and*

$$P(\sigma(z)) = P(z).$$

Then $\sigma(z) = Lz$ for a fixed $L \in \text{GL}(2, \mathbf{C})$ satisfying $P(Lz) = P(z)$ identically.

Proof. After restriction to a small connected open set, polarization of $\|H \circ \sigma\|^2 = \|H\|^2$ gives

$$\langle H(\sigma(z)), H(\sigma(w)) \rangle = \langle H(z), H(w) \rangle.$$

The values of H on any nonempty open set span $\mathbf{C}^3$, since its components are linearly independent polynomials. The same is true of $H \circ \sigma$. Their identical Gram kernels consequently give a constant unitary matrix U with

$$H \circ \sigma = UH.$$

This familiar polarization argument can also be obtained by choosing three points at which the vectors $H(z)$ are independent: their Gram matrix determines an isometry, and inner products with these three vectors determine the equality at every other point.

Write $V(z) = (z_1^2, z_1 z_2, z_2^2)$. Since $H = AV$ for an invertible matrix A , the preceding identity gives

$$V(\sigma(z)) = BV(z), \quad B = A^{-1}UA.$$

Denote the three entries on the right by Q_1, Q_2, Q_3 . They are linearly independent homogeneous quadratics, and

$$Q_1 Q_3 = Q_2^2$$

as a polynomial identity. The polynomials Q_1, Q_3 are relatively prime. Otherwise a common linear factor also divides Q_2 , putting three independent quadratics in the two-dimensional space of quadratics divisible by that factor; a common quadratic factor is even more immediately impossible.

Unique factorization now implies that Q_1, Q_3 are squares of linear forms, after absorbing nonzero constants:

$$Q_1 = \ell_1^2, \quad Q_3 = \ell_2^2, \quad Q_2 = \pm \ell_1 \ell_2.$$

The identities $\sigma_1^2 = Q_1$, $\sigma_2^2 = Q_3$ imply, on a connected open set, that each component of σ equals one of the corresponding linear forms with a fixed sign. Thus $\sigma = Lz$; local invertibility makes L invertible. The norm identity extends polynomially to all of $\mathbf{C}^2$. $\square$

9.3. The isometry group and its smooth quotients. Let

$$K = \{L \in \mathrm{GL}(2, \mathbf{C}) : P(Lz) = P(z)\}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Lemma 9.4. *The full linear isometry group of P is*

$$K = \{\lambda I, \lambda S, \lambda T, \lambda ST : |\lambda| = 1\}. \quad (9.6)$$

Its nonidentity complex reflections are precisely

$$\pm S, \quad \pm T, \quad \pm iST.$$

They generate the group

$$\mathcal{R} = \{i^a I, i^a S, i^a T, i^a ST : 0 \leq a < 4\}, \quad |\mathcal{R}| = 16. \quad (9.7)$$

Proof. The two-sided comparison $c\|z\|^4 \leq P(z) \leq C\|z\|^4$ makes K a compact matrix group. Average a positive definite Hermitian form over K . Both S and T preserve P , so the averaged form is invariant under S and T . Invariance under S makes its matrix diagonal, and invariance under T makes the two diagonal entries equal. Consequently the averaged form is a scalar multiple of the standard one, and $K \subset U(2)$.

Set

$$s = |z_1|^2 + |z_2|^2, \quad x = 2\mathrm{Re}(z_1 \bar{z}_2), \quad y = 2\mathrm{Im}(z_1 \bar{z}_2), \quad t = |z_1|^2 - |z_2|^2.$$

Then $s^2 = x^2 + y^2 + t^2$, and direct expansion gives

$$P = \frac{1}{2}s^2 + \frac{3}{8}x^2 + \frac{1}{8}y^2 + \frac{1}{2}t^2. \quad (9.8)$$

Conjugation of traceless Hermitian 2×2 matrices identifies the projective action of $U(2)$ on (x, y, t) with $SO(3)$; its kernel consists of scalar unitary matrices. The last three coefficients in (9.8) are distinct. An orthogonal transformation preserving this quadratic form must preserve its three individual eigendirections. Its orientation-preserving stabilizer therefore consists of the four diagonal sign changes with determinant one. Their lifts are the four scalar cosets represented by I, S, T, ST . This proves (9.6).

The eigenvalues of S, T are $1, -1$, and those of ST are $i, -i$. A nonscalar element in (9.6) has eigenvalue one precisely at the six matrices displayed in the statement. They all have order two and fix a complex line pointwise. Finally $S^2 = T^2 = I$, $TS = -ST$, and $(ST)^{-1}(iST) = iI$. These identities give exactly the sixteen matrices in (9.7), which are distinct. $\square$

Proposition 9.5. *Let $G : (\mathbf{C}^2, 0) \rightarrow (\mathbf{C}^2, 0)$ be finite. If ϕ is constant on the local fibers of G , then $\deg_0 G \leq 16$.*

Proof. Put $d = \deg_0 G$. Around a generic regular fiber, the d local inverse branches give correspondences σ preserving ϕ , and hence preserving P . By Lemma 9.3 each is the restriction of an element of K . The identities $G \circ \sigma = G$ extend to germ identities $G \circ L = G$. To see this without any domain assumption, take a small Euclidean ball on which G is represented. By Lemma 9.4, L is unitary; thus $G(Lz)$ and $G(z)$ are holomorphic on the same ball, and equality on the nonempty branch open set implies equality throughout the ball.

It follows that

$$\Gamma = \{L \in K : G \circ L = G \text{ as germs}\}$$

is a finite group of order d , and its orbits are the generic fibers of G . Indeed restriction to one regular source open set injects Γ into the d possible inverse branches, and the preceding argument supplies all d branches.

The map therefore identifies its smooth target germ with the quotient germ $(\mathbf{C}^2, 0)/\Gamma$. Here is a ring justification. Put $R = \mathbf{C}\{z_1, z_2\}$ and $B = G^*\mathbf{C}\{w_1, w_2\}$. Then R is finite over the normal ring B , and

$$B \subset R^\Gamma \subset R.$$

The generic degrees give

$$[\text{Frac } R : \text{Frac } B] = d = |\Gamma| = [\text{Frac } R : \text{Frac}(R^\Gamma)].$$

Thus R^Γ is integral over B with the same fraction field, so normality gives $B = R^\Gamma$.

The Chevalley–Shephard–Todd smooth-quotient criterion says that a finite linear group with a smooth quotient at the image of the origin is generated by complex reflections [9, Theorem 1.1 and Corollary 1.8]. It applies here to the faithful finite group Γ . Equivalently, regularity of the local invariant ring implies regularity of the graded invariant ring at its vertex; a homogeneous basis of its cotangent space then generates the invariant polynomial algebra, and the classical polynomial-invariant criterion applies.

Every generating reflection of Γ is consequently one of the six matrices in Lemma 9.4. Hence $\Gamma \subset \mathcal{R}$, and $d = |\Gamma| \leq 16$. $\square$

Proof of Theorem 9.1. The weight hypotheses were verified in Section 9.1. For each fixed j , Lemma 9.2 and Proposition 9.5 give $\deg_0 G_j \leq 16$. The finite-map mass formula [3, Chapter III] gives

$$\tau(m_j^{-1} \log \|G_j\|) = \frac{\deg_0 G_j}{m_j^2}.$$

It tends to zero if $m_j \rightarrow \infty$, proving the asserted contradiction to mass convergence. $\square$

Remark 9.6. The bound also shows that a tower converging to this representative can contain at most four transitions of local degree greater than one: local degrees multiply, and each such transition multiplies the accumulated degree by at least two.

9.4. Arbitrary positive normalization scales.

Corollary 9.7. *Assume convergence in L_{loc}^1 , or locally uniformly off zero. With the Euclidean norm, there is no tower in Theorem 9.1 satisfying $\deg_0 G_k/m_k^2 \rightarrow 1$, even if $m_k \rightarrow \infty$ is not assumed.*

Proof. The bound $1 \leq d_k = \deg_0 G_k \leq 16$ has already been established. Each transition is finite: a positive-dimensional zero fiber of F_k would have a positive-dimensional inverse image under the finite map G_k , contrary to finiteness of G_{k+1} . Local degrees are therefore multiplicative, so d_k is eventually a constant integer d . The mass condition forces $m_k \rightarrow m = \sqrt{d} > 0$.

Convergence in L_{loc}^1 of the psh functions q_k gives local uniform upper bounds, after restriction to a smaller fixed ball. Under local uniform convergence off zero, the maximum principle instead extends upper bounds from a sphere across zero.

Since m_k is bounded, the G_k form a locally bounded family of holomorphic maps. A subsequence converges locally uniformly to a map G . Taking an almost-everywhere convergent subsequence of q_k , if necessary, gives

$$\|G(z)\| = e^{m\phi(z)} = P(z)^{m/4}. \quad (9.9)$$

The equality holds everywhere by continuity. In particular $G^{-1}(0) = \{0\}$ near zero.

We claim that m is an integer $p \geq 1$ and G is homogeneous of degree p . If p is the first nonzero homogeneous degree in the Taylor expansion of G , compare (9.9) at rz , with a generic z , as $r \rightarrow 0$; homogeneity of P gives $m = p$. Write $G = \sum_{\ell \geq p} G_\ell$, and average

$\|G(re^{i\theta}z)\|^2 = r^{2p} \|G(z)\|^2$ over θ . Orthogonality of the powers gives

$$\sum_{\ell \geq p} r^{2\ell} \|G_\ell(z)\|^2 = r^{2p} \|G(z)\|^2.$$

Its nonnegative coefficients force $G_\ell = 0$ for $\ell > p$.

The two homogeneous components of G have no common zero on $\mathbf{P}^1$, so they define a degree- p holomorphic map

$$g : \mathbf{P}^1 \longrightarrow \mathbf{P}^1.$$

Equation (9.9) implies that $g^*\omega_{\text{FS}}$ is a positive constant multiple of the strictly positive form on $\mathbf{P}^1$ induced by ϕ . Consequently g has no critical point. An unramified self-cover of the simply connected sphere has degree one, hence $p = 1$ and $G = Lz$ for an invertible linear map.

It would follow that $P = (z^*L^*Lz)^2$, which is impossible. Indeed, writing the positive definite Hermitian form as $a|z_1|^2 + b|z_2|^2 + cz_1\bar{z}_2 + \bar{c}z_2\bar{z}_1$, the pure fourth-degree terms force $a = b = 1$. The absent terms $z_1^2\bar{z}_1\bar{z}_2$ force $c = 0$. The coefficient of $|z_1z_2|^2$ in the square would then be 2, whereas it is 1 in (9.1). This is the required contradiction. $\square$

10. DYNAMICAL SCOPE AND REALIZATION OF SINGULARITY CLASSES

All inputs are local near 0. The ideals, their polynomial contractions, and the transition identities depend on germs. The Bergman normalization specifies the actual representative recovered on a fixed smaller ball. The maps produced are polynomials, so no successively shrinking domains are needed for their compositions. The chosen representative is used to normalize the sizes of the polynomial generators on a fixed neighborhood.

10.1. What the Li–Xia construction adds. Li–Xia [7] use finite polynomial self-maps

$$T_j(z, w) = \left(\frac{1}{2}w^2, \frac{1}{2}z^2 + \varepsilon_j w\right)$$

of $\mathbf{C}^2$, with rapidly decreasing positive ε_j . For $Q_j = 2^{-j} \log \|T_j \circ \dots \circ T_1\|_\infty$, the local degree is 4^j , so $\tau(Q_j) = 1$, while a surviving linear term gives $\nu(Q_j, 0) = 2^{-j}$. They establish a decreasing sequence after truncation, shrinking cutoff cores, and a separate pointwise estimate for the Lelong number of the limit. This produces an isolated maximal weight u with $\tau(u) = 1$ and $\nu(u, 0) = 0$. The finite self-maps and their preserved local degrees provide structure not furnished by our general varying-model theorem. Our different input is an arbitrarily prescribed target.

There is an exact diagnostic for the distinction. For a zero-Lelong-number target, Skoda integrability gives $\mathcal{J}(tu)_0 = R$ for every $t > 0$. The ideals in case (a) of Theorem 1.1 therefore become $I_k = \mathfrak{m}^{\rho^k}$, whence

$$\tau(\psi_k) = \left(\frac{\rho^k}{m_k}\right)^n \longrightarrow 0.$$

Applied to the Li–Xia weight, the theorem still recovers u pointwise, in L^1_{loc} , and locally uniformly off 0, but not by convergence of the masses of the atoms at 0. This does not preclude concentration of non-atomic mass. It shows why function convergence and divisorial convergence alone are inadequate substitutes for the mass estimates in Theorems 1.3 and 8.3. The Li–Xia transitions have linear terms, so Proposition 1.2 does not constrain their scale. Theorem 7.1 now supplies its own parameter-dependent orbit construction on the fixed space $\mathbf{C}^n$, including the zero endpoint in the proof. Its perturbation-and-sweeping mechanism is an adaptation of [7], with variable degrees and a prescribed order ratio. Theorem 8.3 then addresses the mass loss for its stated class of backgrounds by retaining the zero-endpoint orbit ideals as a separate factor.

10.2. Exact representatives and singularity classes. For the arbitrary prescribed-germ towers, our diagram is

$$\begin{array}{ccc} \mathbf{C}^n & \xrightarrow{\text{id}} & \mathbf{C}^n \\ H_k \downarrow & & \downarrow H_{k+1} \\ X_k & \xrightarrow{F_k} & X_{k+1}. \end{array}$$

Thus high-order vanishing is measured in successive ambient embeddings; it is not contraction on the common normalization. Neither a uniform bound on N_k nor invariance of one smooth state space is established for these general targets. Numerical pairs do admit the fixed-space construction of Theorem 7.1.

Theorem 9.1 rules out a universal fixed-smooth-space construction satisfying

$$G_{k+1} = F_k \circ G_k, \quad m_k^{-1} \log \|G_k\| \longrightarrow \phi, \quad \frac{\deg_0 G_k}{m_k^n} \longrightarrow \tau(\phi)$$

for every prescribed tame maximal representative, with $m_k \rightarrow \infty$ and convergence on a fixed neighborhood. Corollary 9.7 excludes arbitrary positive scales as well for its Euclidean-norm, L^1_{loc} formulation.

The obstruction does not apply to realization modulo $O(1)$. For the explicit weight in Section 9, the coordinate-power tower

$$G_k(z) = (z_1^{2^k}, z_2^{2^k}), \quad F_k(w) = (w_1^2, w_2^2), \quad m_k = 2^k$$

has normalized mass one at every stage and converges to $\log \max(|z_1|, |z_2|)$, a different representative of the same singularity class. Thus the counterexample separates exact representative prescription from prescription of a class. It is also compatible with Theorem 1.1: the latter’s intermediate models need not be smooth, whereas smoothness is precisely what forces reflection generation in Proposition 9.5.

A remaining question is whether every tame maximal singularity class admits some representative realized by finite towers on fixed $(\mathbf{C}^n, 0)$, with convergent atomic masses. Generic minimal reductions supply n -component finite maps with the correct singularity at each individual level [8, Proposition 3.3], but do not ensure successive holomorphic factorization. Containment in powers of the preceding ideal does not imply membership in the algebra generated by an arbitrary minimal reduction. The algebras $\mathbf{C} + I_k$ provide this membership on varying models; the class-level fixed-space question is not settled by the present results.

Global automorphisms would impose a further change of mechanism: their local degree is one, so the normalized logarithms have residual mass m_k^{-n} at every finite stage. The uniformly attracting automorphism-basin problem associated with Bedford’s conjecture is already settled by Bera–Verma [2]; it does not turn the present finite, potentially branched realization problem into an automorphism construction.

Declaration on the use of artificial intelligence. This working manuscript was prepared with AI assistance. The fixed-smooth-space counterexample and the present revision were developed with OpenAI Codex (GPT-6). Mathematical correctness and literature novelty require independent verification.

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